

emissions which use the same spectrum, the other devices can effectively “jam” your device’s transmissions.

To avoid interference, which seems inevitable considering the host of devices using the same band, Bluetooth radios utilise a technique called spread spectrum frequency hopping (also known as frequency-hopping spread spectrum, FHSS). This is a method of transmitting radio signals by rapidly switching a carrier among many frequency channels. Splitting the signal across various frequencies as opposed

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to keeping it on a fixed one allows for three major advantages. One, it reduces narrowband interference. During the recollection process of the spread signal, the interfering signal is also spread, which allows the receiver to eliminate it as it disappears into the background. Two, spread signals are difficult to intercept considering they are scattered across various frequencies in a pseudo-random order. For any eavesdropper, it would appear as noise unless the exact order of scattering of the signal was known. Three, a spread spectrum transmission can share a frequency band with most other signals with minimal interference. It also adds minimal noise to narrow frequency communications. This improves the efficiency of bandwidth utilisation.

For easy transmission of data, to avoid large losses, data is typically broken up into very small parts, called packets. FHSS isn’t a new technology, but Bluetooth is more efficient than most other devices utilising the spectrum since the devices typically hop faster and use shorter packets. All Bluetooth devices are capable of transmitting both voice and data signals.

Any two Bluetooth enabled devices that come within 30 feet of each other can set up an ad hoc point to point or point to multipoint connection. So, as you travel, your personal network travels with you. There are two major states of operation in a Bluetooth device, connection and standby. If a device is connected and not actively involved with current activities, then standby mode is triggered. This acts as a power conservation